ON THE PHYSICAL BASIS OF INFORMATION FIDELITY

A Unified Theory of Information Integrity, Its Degradation, and Its Measurement

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Consolidating:
Training Data Velocity Bias (Zenodo, 2025; doi:10.5281/zenodo.17459755)
Vital Network Science (Zenodo, 2025; doi:10.5281/zenodo.17459844)
Photonic Substrate Intelligence (Zenodo, 2026; doi:10.5281/zenodo.18995268)

ABSTRACT

Information has no metrology. Every other quantity exchanged between humans acquired standardised measurement. Information did not. The result is a compounding integrity loss across every retransmission and every political cycle, with no instrument capable of detecting it.

This monograph unifies three prior results into a single theory. Training Data Velocity Bias (2025) demonstrated that information degrades through retransmission, with false content spreading faster than true, contaminating AI training corpora. Vital Network Science (2025) proposed structural intervention: regulating information velocity rather than content, treating information ecosystems as control systems requiring homeostatic governance. Photonic Substrate Intelligence (2026) described a computing architecture using light rather than electronics, with Orbital Angular Momentum modes as orthogonal channels and thermomagnetic tape as physical memory.

This monograph reveals that all three investigated different faces of the same absence. It makes three claims. First, that the Colour Rendering Index provides a structural precedent for measuring information fidelity, down to the anchoring of its scale by convention. Second, that the PSI architecture provides the physical instrument. Third, that light is the natural physical realisation of information fidelity: interpretive transmission and optical decoherence are channels governed by one mathematics, and in the optical channel the loss is directly readable as fringe visibility, a measurement optics has performed since Young announced the interference principle in November 1801.

We define the Information Rendering Index (IRI), a unit of information fidelity (Ω, the render), and a complete metrological chain: two declared measurands, artefact integrity and content fidelity, separated by a versioned canonicalisation transducer; a three-tier reference hierarchy with stated uncertainties under JCGM 100; an anchoring convention that fixes the scale; and a falsification-first experimental programme. Every certified result is reported as Ia $\pm$ U, with the budget on the record. The prior notation is harmonised: φ (virality) and ψ (vitality) from the earlier papers are preserved; the Greek letter Ψ is not used for photonic states, which are denoted $|\mathrm{I}\rangle$ and $|\mathrm{R}\rangle$ to avoid collision with the PSI architecture acronym.

PART ONE: Principle

1. Statement

Interpretive transmission loses information. The data processing inequality makes this a theorem: any channel that works from a derivative alone, without returning to the source, preserves at best and never increases what the derivative carries about that source. Optical decoherence obeys the same channel mathematics, and optics presents the loss as a directly readable number, fringe visibility. A defined quantity whose governing mathematics appears in a physical system in measurable form can be realised physically; this is the founding pattern of metrology. Information fidelity is such a quantity. This monograph defines it, states its axioms, and identifies its natural physical realisation in light.

2. Axioms

A1. Information fidelity is a definable quantity: bounded, with a fixed value on perfect transmission.

A2. Its measurement must be grounded in physics to escape the recursion problem. An instrument built from interpretation inherits the defect it measures.

A3. Fidelity is monotone under interpretation: composition with any further interpretive channel leaves the index unchanged at best. This is the data processing inequality in operational form.

A4. Degradation is continuous in the strength of interpretive extraction, and in a memoryless cascade it compounds multiplicatively.

The remainder of the monograph builds the realisation. Part Nine states these axioms formally and positions the index they define against the established measures of information.

3. Notation

This monograph harmonises notation across four texts. Conventions:

$\varphi(c,t)$: virality; cumulative exposure of content c at time t. From VB and VNS.

$\dot{\varphi}(c)$: velocity; rate of spread $d\varphi/dt$. From VB and VNS.

$\psi(c)$: vitality; epistemic quality of content c. From VB. In VNS, $\psi(t)$ is the composite index.

$\beta\varphi$: velocity bias coefficient. From VB.

$\rho\varphi,\psi$: velocity-accuracy anticorrelation. From VB.

G: vitality governor function. From VNS.

g: governor feedback gain. From VNS, where it appeared as k; renamed here to reserve k for the coverage factor of the uncertainty doctrine (Part Ten).

$|I\rangle$: information state (photonic). New notation replacing $|\Psi_{info}\rangle$ to avoid collision with the PSI acronym and the ψ variable.

$|R\rangle$: reference state (photonic). New notation replacing $|\Psi_{ref}\rangle$.

Ω: the render; unit of information fidelity. New.

Ia: Information Rendering Index (0-100). New.

ℓ: OAM topological charge. From PSI.

κ: scaling constant of the index (Part Six). Γ: decoherence rate. τ_m: mode measurement time. N: dimension and mode count.

T (subscript v): canonicalisation transducer, version v; the declared procedure projecting content into canonical physical form. New.

$|I_c\rangle$, $|R_c\rangle$: canonical content states; the photonic states produced by illuminating or projecting the canonical renderings of derivative and source. New.

V_A: artefact match visibility; correlation of an object's current surface state against its own recorded hologram. Reported separately from Ia. New.

V_c: content match visibility; global visibility of the interference between the canonical content states of source and derivative. New.

u_{inst}, u_T, $u(R)$: uncertainty components of instrument, transducer, and reference. u_c: combined standard uncertainty. U: expanded uncertainty. k: coverage factor (JCGM 100). New.

$s(R)$, n_{eff}, η_i, u_w, u_{cal}, u_{com}: Tier 2 reference budget quantities; constituent dispersion, effective independent constituent count, eigenvalues of the constituent correlation matrix, weighting sensitivity, calibration residual, common-origin allowance. New.

PART TWO: The Absence

4. What does not exist

Shannon (1948) defined the bit and channel capacity. He excluded semantics. [1]

CRI, introduced in 1965 and currently specified in CIE 13.3-1995, measured rendering fidelity of light, and of light only. [2]

Information Quality frameworks (Wang & Strong, 1996) produced subjective taxonomies without physical referents. [3]

The Admiralty Code formalised expert judgement without grounding it in physics. [4]

No measurement exists of how faithfully information renders reality.

5. The compounding error

Dionysius of Halicarnassus imposed Athenian categories onto Roman institutions. Livy propagated different distortions. Cary identified both while introducing his own. Each believed he was correcting the record. [5]

On a four-year political cycle, the same mechanism operates at a pace that precludes correction. Every existing intervention, journalism, fact-checking, peer review, AI, is an interpretive layer, and an interpretive layer is a channel (Section 24). Without a ruler, the rope is shaved indefinitely.

PART THREE: The Symptom

from Training Data Velocity Bias (2025)

6. Velocity bias

False information spreads faster, farther, and more broadly than truth. Vosoughi, Roy, and Aral (2018) demonstrated across 126,000 news cascades that falsehoods were 70% more likely to be retransmitted than accurate content. [6]

Training corpora built from these ecosystems inherit this asymmetry. The velocity bias coefficient, where w(c) is the corpus weight of content c:

$$\beta\varphi = E\big[w(c) \mid \dot{\varphi}(c) > \theta_{\text{high}}\big] / E[w(c) \mid \dot{\varphi}(c) < \theta_{\text{low}}] \tag{1}$$

When $\beta\varphi > 1$, high-velocity content is overrepresented. When velocity anticorrelates with accuracy ($\rho\varphi,\psi < 0$), the corpus systematically overrepresents falsehood.

7. From velocity bias to hallucination

Models trained on velocity-biased corpora learn that rapid spread correlates with truth. They internalise viral patterns as validity signals. Hallucination is the model faithfully reproducing the statistical properties of its training data. [7]

Kalai and Vempala (2024) proved that calibrated language models must hallucinate on arbitrary facts, those whose veracity cannot be determined from the training data, at a rate lower-bounded by the monofact rate: the fraction of such facts appearing exactly once in the training set, minus a calibration term. [8] The bound follows from a Good-Turing missing-mass argument: content that appears once is exactly the content whose next-token probability the model cannot confidently distinguish from novel fabrication. Rarely corroborated content is precisely what velocity-biased corpora overproduce: the same ecosystems that amplify viral content generate novel claims faster than corroboration can consolidate them, so each novel claim appears once or in slightly mutated variants that fail to cluster. The monofact rate of a φ-biased corpus is therefore inflated by the same dynamics that inflate $\beta\varphi$; this is a testable hypothesis, and Experiment 3.1 measures the monofact rate in velocity-biased and velocity-normalised samples directly. The theorem connects velocity bias to hallucination through the statistics of the corpus itself.

The training distribution distortion:

$$P_{\text{train}}(c) / P_{\text{true}}(c) \propto \varphi(c)^{\alpha} \qquad \alpha > 0 \tag{2}$$

When $\alpha > 0$ and $\rho\varphi,\psi < 0$, training data systematically overrepresents inaccurate content.

8. The feedback loop

As AI-generated content enters future training corpora, velocity bias compounds through recursion: models trained on φ-biased data produce φ-biased output, which contaminates the next training cycle. Each generation inherits and amplifies the distortion of the last.

This is cascade degradation at civilisational scale. The contribution of VB is to have measured the symptom of a channel process without yet identifying the governing mathematics; Part Nine supplies it and names its physical realisation.

PART FOUR: The Treatment

from Vital Network Science (2025)

9. The vitality index

VNS formalised vitality ψ(t) as a composite of seven dimensions: coherence (C), resilience (R), civility (V), epistemic quality (Q), diversity (D), agency (A), and affective balance (B). [9]

$$\psi(t) = \Sigma_{i=1}^{7} w_i X_i(t) \qquad X_i \in [0,1], \Sigma w_i = 1 \tag{3}$$

Each component captures a distinct aspect of informational health, independently measurable and independently degradable.

10. The governor

VNS proposed a proportional-derivative controller monitoring velocity excursions:

$$G(\varphi, \dot{\varphi}) = 0 \text{ if } \dot{\varphi} \leq \theta; g(\dot{\varphi} - \theta) \text{ if } \dot{\varphi} > \theta \tag{4}$$

Where θ is the velocity threshold and g is feedback gain. Intervention is temporal (seconds to minutes), and never censorial. The principle: tune time, and leave truth alone.

11. The key insight from VNS

VNS identified that intervention must be structural, and structural in this context means physical. Content moderation is another Dionysius, another layer of confident interpretation. Velocity regulation targets the transmission rate, and the transmission rate alone. This is the correct instinct: the loss is channel loss compounding at cascade speed, so the intervention addresses the rate at which interpretive layers accumulate.

What VNS lacked was the physical grounding. The ψ index was defined as a composite of measured social indicators, still, ultimately, interpreted. ψ names a measurable quantity, and Part Nine supplies the measurement. The seven VNS dimensions map onto the IRI test dimensions (Section 19), and the governor G triggers on a physically measurable quantity: Ω-rate.

PART FIVE: The Instrument

from Photonic Substrate Intelligence (2026)

12. The compiler

Every existing information system is interpreted. Content passes through layer after layer of software or editorial process, each adding its own bias. In software terms, this is an interpreted language: executed line by line, overhead accumulating at every step.

PSI compiles information directly into physics. The architecture comprises an optical inference stack (matrix multiplication at the speed of light, in the lineage of diffractive deep neural networks demonstrated by Lin et al. [12]), OAM-encoded gradient channels (orthogonal physical modes for independent data streams), and a thermomagnetic tape substrate (physical, non-volatile, tamper-evident weight storage). [10] The training of the optical stack itself can proceed by in situ backpropagation through the physical system, measuring the gradient optically rather than computing it digitally (Hughes et al., 2018 [42]).

For artefact integrity, the measurement operates on the raw substrate with no interpretive layer between light and matter. For content fidelity, one declared transduction step (Part Ten) enters the chain, and that step is frozen, versioned, and recorded.

13. OAM modes as fidelity channels

Orbital Angular Momentum modes are characterised by topological charge ℓ, each carrying $\ell\hbar$ of angular momentum per photon [38]. Each mode is orthogonal to every other. Wang et al. (2012) demonstrated terabit-per-second data rates using OAM multiplexing in a laboratory free-space link [11].

In the PSI architecture, OAM modes carry gradient information during training. In the IRI application, each mode is assigned to one dimension of information fidelity (D1-D14). The orthogonality guarantee means fidelity measurement on one dimension cannot contaminate another, unless distortion introduces cross-modal coupling, which is physically detectable.

14. The thermomagnetic calibration record

Physical metrology demands traceability. The PSI thermomagnetic tape records the training process as a physical state: auditable, tamper-evident, temporally anchored. It is the calibration certificate of the measurement system. The substrate belongs to a broader class: Li et al. (2025) demonstrated a compact cassette tape for DNA-based data storage, establishing that physically addressable recording substrates now exist across multiple material platforms [39]. Every reference state, every transducer version, every uncertainty budget (Part Ten) is recorded on this tape, so that a measurement can be reconstructed from its record and a certificate that cannot be reconstructed is void.

15. What PSI is

PSI was conceived as a compute architecture: an optical core of roughly 27 litres, a 30 cm cube, running tens of thousands of concurrent frontier-class inferences via WDM and a 100×100 stack array at kilowatt-scale power. [10]

The cube is the computing volume, and the honest envelope is larger. Illumination drivers, the detector readout, the tape decks, and above all the rejection of tens of kilowatts of heat place the installed instrument at equipment-rack to small-vehicle scale, with the thermal plant the dominant term. The comparison that carries the architecture survives this accounting and

sharpens under it: the electronic equivalent of the same concurrent capacity draws megawatts in a purpose-built building, and the photonic system draws kilowatts in a vehicle.

That remains true at system scale. PSI's deeper identity is metrological: the first physical substrate on which information integrity can be measured rather than interpreted.

PART SIX: The Information Rendering Index

16. The structure inherited from CRI

CRI measures how faithfully a test light source renders colours against a reference illuminant: fidelity of rendering, scored on a 0-100 scale. [2]

$$Ra = (1/N)\Sigma_i R_i \qquad R_i = 100 - 4.6 \times \Delta E_i \tag{5}$$

The IRI retains this structure:

$$Ia = (1/N)\Sigma_i I_i \qquad I_i = 100 - \kappa \times \Delta\Omega_i \tag{6}$$

$$\Delta\Omega_i = \sqrt{[(\Delta p_i)^2 + (\Delta t_i)^2 + (\Delta c_i)^2]} \tag{7}$$

where Ia is the IRI (0-100), $\Delta\Omega_i$ is deviation per dimension in Information Fidelity Space (axes: p, provenance; t, temporality; c, coherence), and κ is the scaling constant of the index. The three axes enter equation (7) with equal weight; this equal weighting is a retained convention, and a complete derivation of the fidelity-space geometry must either produce the normalisation or declare it retained. The monograph declares it retained (Open Questions, item 1).

The inherited structure carries a history, and the history settles a question this monograph must answer about κ. The 4.6 in equation (5) was never derived from the physics of light or the physiology of vision. The designers of the CRI fixed it by convention: the factor was selected so that a standard warm white halophosphate fluorescent lamp would score Ra = 51. [13, 14] Every successor index renewed the same move. The Colour Quality Scale set its factor of 3.1 so that the mean score of the CIE F1 to F12 fluorescent series would equal the mean Ra for the same lamps. [13] IES TM-30-15 set its factor of 7.54 by equalising its mean fidelity value with mean Ra across the CIE F-series illuminants. CIE 224:2017, the index the CIE designates for accurate scientific use, revised the factor to 6.73, chosen to equalise mean Rf with mean Ra across a library of commercially available sources. [15, 16]

Four generations of colour fidelity metrology, and each one anchored its scale to a conventional artefact set.

The lesson generalises. In an index of this family, the deviation is physical and the scale is conventional. The colour difference ΔE_i is a measured quantity; the number of index points one unit of difference costs is a published, versioned decision that makes scores comparable across instruments and decades. The convention is the mechanism by which a physical measurand becomes a shared scale.

The IRI adopts the discipline together with the algebra: $\Delta\Omega_i$ is the physical measurand, κ is the scale constant, and κ is fixed by a stated anchoring convention, defined in the next section.

17. The anchoring convention

The render scale is anchored to a conventional reference transformation.

Anchoring convention (Ω-90). One professional single-pass summarisation layer scores 90 Ω. Formally: the panel mean index of the frozen anchor set A equals 90, which fixes the scaling constant of equation (6):

$$\kappa = 10/\langle\Delta\Omega\rangle_A \tag{8}$$

where $\langle \Delta\Omega \rangle_{\mathrm{A}}$ is the grand mean deviation of the anchor set across artefacts and dimensions, measured at reference Tier 1 under the versioned canonical-rendering transducer (Part Ten).

The anchor set A is specified with the experimental programme: the fifty professional summaries of Experiment 1.1, degradation class (d), each produced by a trained analyst working from the source document alone under the written summarisation protocol, frozen together with their source documents at a stated transducer version and held on the thermomagnetic calibration record. The anchor plays the role the warm white halophosphate lamp played for the CRI: a defined artefact set, standing still, against which the scale is set. [13]

The anchor is measured, and never judged. Its artefacts are produced by humans; the convention consumes their measured deviations and nothing else. No assessor scores the anchor summaries, no rubric grades them, and no editorial judgement enters the chain. The CRI's anchor lamp was likewise a manufactured object, and its manufacture by humans never made the measurement of its spectrum interpretive.

What the convention fixes is one point of the scale. Perfect transmission scores 100 by the identity axiom; the conventional reference transformation scores 90 by decision; linearity in $\Delta\Omega$, equation (6), extends the scale between and below these points. A fixed ceiling and one anchor determine κ completely.

What the convention leaves open is everything empirical. It leaves the measurand untouched: $\Delta\Omega_{\mathrm{i}}$ remains a physical deviation between photonic states. It leaves the composition law untouched: whether degradation composes multiplicatively across layers is an empirical claim, tested by Experiment 1.3, and no choice of κ can rescue or condemn it. It assigns no score to any artefact class beyond the anchor: journalistic layers, translations, and social retransmission score wherever measurement puts them. A different anchor value would regauge the scale linearly, changing no ordering, no ratio of losses, and no experimental verdict.

The choice of 90 aligns the index with the cascade vocabulary. Under the instrument form of the index, equation (13), a score of 90 corresponds to a mean mode-power ratio of 0.90, so the anchor reads in cascade terms as the assignment of per-layer coefficient 0.90 to the conventional reference transformation: the first row of the cascade table in Section 25. The alignment holds at the anchor point by construction; measured coefficients for every other layer class are Experiment 1.3′s to deliver.

One further repair completes the scale. The linear form of equation (6) is unbounded below, so a catastrophic deviation can drive the index negative, in conflict with the boundedness axiom. The successor colour indices met the same defect with a fixed transform, mapping the linear score through a soft floor; the CQS and ANSI/IES TM-30-20 both use the same device. [13, 17]

$$Ia = 10 \ln[\exp(I'a/10) + 1] \tag{9}$$

applied to the general index after averaging and to each per-dimension index individually, where I'a is the linear value from equation (6). The transform, followed by the reporting cap at 100, carries $(-\infty, 100]$ onto $(0, 100]$. It moves a score of 90 by about one thousandth of a point and a score of 50 by less than a tenth of a point, and confines its effect to the catastrophic end of the scale, which it compresses into the interval above zero. Scores are reported to one decimal place and capped at 100.

A convention earns its authority through versioning. The anchor set, the summarisation protocol, the transducer version, the reference tier, and the resulting value of κ are recorded

together on the calibration record; any revision, whether a new transducer version, an enlarged anchor panel, or a deliberate re-anchoring, produces a named scale version, exactly as CIE 224:2017 is a named re-anchoring of the CIE 13.3 scale. [15] Every reported score carries its scale version.

The scale is anchored the way every fidelity scale in its lineage has been anchored: to a named artefact set, by a published decision, on a record anyone can audit.

18. Test dimensions

	Dimension	Measures	Detects	Maps to VNS
D1	Provenance	Source traceability	Fabrication	Q (Epistemic)
D2	Temporal Accuracy	Time-bound claims	Anachronism	
D3	Internal Coherence	Self-consistency	Contradiction	C (Coherence)
D4	Source Plurality	Corroboration	Single-source	Q (Epistemic)
D5	Contextual Completeness	Framing present	Cherry-picking	D (Diversity)
D6	Spectral Balance	Perspectives	Bias, omission	D (Diversity)
D7	Transmission Fidelity	Retransmission loss	Editorial drift	R (Resilience)
D8	Referential Accuracy	Citation verifiability	Misattribution	Q (Epistemic)

Extended dimensions D9-D14: adversarial resilience (maps to VNS B, affective balance), temporal stability (VNS R, resilience), cross-domain consistency, predictive validity, falsifiability (VNS Q), and metadata integrity (VNS A, agency).

PART SEVEN: Physical Measurement

19. The VNS-IRI mapping

Six of the seven VNS vitality dimensions map onto one or more IRI test dimensions; civility (V) has no direct counterpart in the current D1-D14 set and stands as a candidate for an extended dimension. VNS measured ψ as a composite social indicator. The VNS-IRI mapping and Experiment 3.2 test whether ψ and Ia track the same underlying quantity, one socially observed and the other physically measured. The six mapped VNS dimensions are the social projections of their corresponding IRI dimensions, viewed from a different angle. The vitality index ψ(t) is the socially observed shadow of the physically measured Ia.

20. State encoding

The information state |I⟩ and reference state |R⟩:

$$|I\rangle = \Sigma_i \alpha_i e^{i\ell_i\varphi}|\ell_i\rangle \tag{10}$$

$$|R\rangle = \Sigma_i \rho_i e^{i\ell_i\varphi}|\ell_i\rangle \tag{11}$$

21. Mode fidelity

$$M_i = \min\Big(|\alpha_i|^2/|\rho_i|^2, 1\Big) \tag{12}$$

When the derivative concentrates more power into a mode than the reference, the ratio exceeds unity; the index clips each ratio at 1 before averaging, so that no single mode's gain masks another mode's loss.

22. The core equation

The index of equation (6) is defined in Information Fidelity Space. The instrument realises it in the OAM mode basis:

$$Ia = (1/N)\Sigma_i \Big[100 \times \min\Big(|\langle\ell_i|I\rangle|^2/|\langle\ell_i|R\rangle|^2, 1\Big)\Big] \tag{13}$$

Equation (13) is the instrument form of the index: the mode-basis realisation of equation (6), with each mode fidelity $M_i = |\langle\ell_i|I\rangle|^2 \,/\, |\langle\ell_i|R\rangle|^2$ replacing the corresponding dimension score I_i. The two forms coincide at the anchor point by construction; their general relationship, including the conditions under which (13) is an unbiased estimator of (6), is an open question (Open Questions).

23. Distortion diagnostics

Amplitude reduction: information loss. Mode coupling ($C_{ij} = |\langle\ell_i|D_{ch}|\ell_j\rangle|^2$, where D_{ch} is the channel's distortion operator, distinct from the distinguishability D of Section 29): cross-dimensional contamination. Phase shift: systematic bias. Together these form the distortion fingerprint.

In VB terms: a velocity-biased training corpus shows characteristic amplitude reduction on D1 (Provenance) and D4 (Source Plurality), with mode coupling between D7 (Transmission) and D6 (Balance). In VNS terms: the governor G triggers when the Ω-rate of a content cascade exceeds the physically calibrated threshold.

PART EIGHT: Cascade Degradation

24. The interpretive channel

Each retelling in a transmission chain works from the account before it. The journalist writes from the wire copy, the wire copy condenses the release, the release summarises the report, and the reader's recollection compresses the journalist. Nobody consults the original. Formally, the chain $\mathrm{S} \to \mathrm{Y}_1 \to \mathrm{Y}_2 \to \ldots \to \mathrm{Y}_\mathrm{L}$ is Markov: every stage is a stochastic function of its predecessor alone.

For any such chain, the data processing inequality states:

$$I(S; Y_{j+1}) \leq I(S; Y_j) \tag{14}$$

where I is Shannon mutual information [18, Theorem 2.8.1]. No processing of a derivative alone can create information about the source; the inequality allows preservation at best, and every real editorial layer falls short of it. The theorem holds for every channel without exception: careful summary, sloppy paraphrase, faithful translation, deliberate distortion. Monotone degradation under interpretation is a proved property of transmission chains, and the cascade table of the next section is a model of its rate.

The theorem carries an operational corollary. A later stage can appear to repair a distortion only by importing side information, and side information from anywhere except the source merely splices in a second chain with losses of its own. The single act that restores fidelity is a return to the source, which starts the count of layers again from zero. The data processing inequality is the theorem that says a fact-checker must consult the primary.

25. The multiplicative law

The data processing inequality fixes the direction of degradation and leaves the functional form open. The multiplicative law is the form degradation takes when the layers are memoryless and mutually independent, and two standard results show why.

In the simplest discrete channel, the binary symmetric channel with error probability p, the surviving correlation between input and output is 1 - 2p. Compose two such channels and the total error probability is $\mathrm{p} = \mathrm{p}_1(1 - \mathrm{p}_2) + \mathrm{p}_2(1 - \mathrm{p}_1)$, so that:

$$1 - 2p = (1 - 2p_1)(1 - 2p_2) \tag{15}$$

The fidelity parameter of the composite channel is the product of the parameters of its stages, exactly. In optics the same law appears as coherence: a beam traversing a sequence of independent decohering elements retains a fringe visibility equal to the product of the factors the elements leave [19], and the exponential decay of coherence with time is the continuous limit of the same product.

The cascade law adopts this form for interpretive layers:

$$M_i(\text{total}) = \prod_j M_{ij} \qquad j = 1 \text{ to } L \tag{16}$$

Layers	Per-Layer	Cumulative	Remaining	Equivalent to
1	0.90	0.900	90%	Single citation
5	0.90	0.590	59%	News cycle
10	0.90	0.349	35%	Viral cascade
20	0.90	0.122	12%	Political cycle
50	0.90	0.005	<1%	Historiography

The table's per-layer coefficient of 0.90 restates the anchoring convention of Part Six in cascade form: the conventional reference transformation, one professional single-pass summarisation layer, sits at mode fidelity 0.90 by decision, and the table's rows are its powers. The rows therefore illustrate the composition law at the anchor value. The empirical content of the table is the multiplicative form itself, which Experiment 1.3 tests against the additive alternative; measured per-layer coefficients for real layer classes are predicted to fall between 0.85 and 0.95 and to vary by document type.

The law is a model, and it is stated as one. The inequality (14) guarantees monotone loss under any composition; multiplicativity is the additional structure of a memoryless cascade, and Experiment 1.3 tests it against the additive alternative on real transmission chains. A cascade that rejects the product form would revise the table and leave the theorem standing.

26. Velocity as cascade accelerator

Vosoughi's finding that false news reaches 1,500 people about six times faster means false content passes through more interpretive layers per unit time. [6]

VB's velocity bias is cascade degradation at speed. VNS's governor G is a cascade rate-limiter. The Ω-rate is the quantity both papers were measuring without the physical framework:

$$\Omega\text{-rate} = d(Ia)/dt \tag{17}$$

The per-mode form, $d(M_i)/dt$, resolves the aggregate rate into its dimensional components. VNS's circuit breaker triggers when Ω-rate exceeds a threshold. On the instrument of Parts Five and Ten the threshold becomes a measured quantity, and Experiment 3.2 tests whether exceeding it predicts the misinformation it exists to catch.

PART NINE: The Realisation

27. The double-slit experiment

When no which-path information is extracted, the photon exhibits full interference. When an observer measures which slit the photon traverses, the interference pattern degrades. [19]

$$V = (I_{\max} - I_{\min})/(I_{\max} + I_{\min}) \tag{18}$$

V = 1: perfect coherence. V = 0: total decoherence.

Fringe visibility is a bounded, dimensionless index, read directly off the detector plane, and the trade between it and the information extracted is exactly quantified (Section 29). Young announced the principle of interference to the Royal Society in November 1801 [20] and demonstrated the interference law experimentally in the Bakerian Lecture of 1803 [21]. Optics has been reading a fidelity index off a screen for more than two centuries.

28. The realisation claim

Metrology has always separated a quantity from its realisation. The metre was defined as a length. From 1889 the definition was realised by a platinum-iridium bar in Sèvres; from 1960 by counting interference fringes of krypton-86 radiation; from 1983 by the fixed value of the speed of light. [22] The bar stood as the stable physical embodiment against which every other length could be compared, and that is precisely what a realisation is: a physical system in which a defined quantity appears in measurable form. For twenty-three years the realisation of the metre was fringe counting. Interferometry has already carried the full weight of a primary standard.

Information fidelity has the same structure. The quantity is defined by the axioms of Section 31: bounded, perfect on exact transmission, monotone under interpretive processing, continuous in the strength of extraction. Interpretive transmission degrades it by theorem (Section 24). The definition owes nothing to optics.

Light is where the defining mathematics appears in directly measurable form. A physical channel contracts distinguishability in the same monotone way that an interpretive channel contracts mutual information: under any physical channel, no pair of states becomes more distinguishable, trace distance is non-increasing and fidelity between the pair is non-decreasing [23, ch. 9]; under any interpretive channel, information about the source obeys the inequality (14). One monotone structure governs both, and the optical channel alone hands the experimenter the loss as a number on a screen. Fringe visibility is bounded on [0, 1], equal to unity on undisturbed transmission, driven down by decohering elements, and continuous in measurement strength. Visibility satisfies the fidelity axioms as a physical observable.

This is the claim of the monograph, stated at its exact strength. Light realises information fidelity the way the bar realised the metre: a stable physical system in which the defined quantity is directly measurable, available for calibration and for comparison. The photonic substrate of Part Five is the analogue instrument built on that realisation. Semantic degradation happens in meaning space, between claims and their sources; optical decoherence happens in a laser and a pair of slits; the correspondence between them is a shared channel mathematics, made visible on one side.

Two experiments police the boundary of the claim. Experiment 1.2 tests whether the semantic degradation curve and the optical decoherence curve share a functional form. Experiment 3.3 tests whether the scale constant κ is derivable from the substrate's decoherence rate,

which would promote the correspondence from shared mathematics to generated scale. Until those results are in, the claim stops where the evidence stops.

What survives at full strength is this. Light is where the universe makes fidelity loss directly visible. The double-slit experiment is the oldest fidelity measurement: a bounded index of coherence, read as fringe visibility, degraded by quantified acts of extraction. Two centuries of interferometry are two centuries of practice at exactly the measurement this monograph requires.

29. Englert's inequality: the index is continuous

A fidelity index that jumped between discrete values would be a classifier. A measurement needs a gradient, and the gradient is established physics. Englert (1996) proved the duality relation [24]:

$$D^2 + V^2 \leq 1 \tag{19}$$

where D is distinguishability, the amount of which-path information extracted, and V is fringe visibility, the coherence the system retains. As D rises, the ceiling on V falls, continuously, through every intermediate value. Partial extraction yields partial decoherence. The transition is a dial, and the dial has every position between zero and one.

The relation is an inequality, saturated when every state in the interaction is pure and strict when part of the record leaks into degrees of freedom nobody tracks; Section 30 does that accounting. What matters here is continuity. D maps to the strength of interpretive extraction. V maps to the fidelity remaining. The IRI's 0 to 100 scale is a scaled continuum, and Englert's relation is the physical warrant that the underlying quantity moves on a continuum at all. Experiment 1.2 implements the mapping directly, with a which-path interferometer of continuously variable measurement strength on one bench and a graded series of document derivatives on the other.

An instrument is calibrated against a known curve. For this instrument, the duality relation is the curve.

30. The fidelity ledger

The inequality reads as a budget. One unit of duality is available; whatever share the observer takes as which-path knowledge, D^2, is unavailable to the system as coherence, V^2. When the photon, the path marker, and everything they touch are described in full, the accounting closes: the joint state evolves unitarily, and coherence lost by the system is stored, exactly, in its correlations with the marker [23]. Equality in (19) is the closed ledger. The strict inequality opens when part of the record escapes into an environment nobody monitors: the books still balance, and some entries sit in an account no observer holds.

This reframes what measurement does. Observation transfers coherence into correlation at a stated price, and the price is visible on the screen as lost visibility. The fringes fade because the fidelity went somewhere definite: into the observer's record, or into the untracked environment. Complementarity, read this way, is bookkeeping for information: a conservation law in the closed description, an inequality wherever the description is cut short.

For the IRI the lesson is architectural. A measurement that keeps its records keeps the ledger auditable; the thermomagnetic tape of Part Five exists so that the instrument's own extractions stay on the books. Fidelity that leaves the system must land somewhere the metrology can see.

31. The axioms of a fidelity index

The monograph's definitional content compresses to four requirements. An information fidelity index is a functional $\Phi(R, I)$ on a reference R and a derivative I, satisfying:

F1. Boundedness. Φ takes values in [0, 100].

F2. Identity. $\Phi(R, I) = 100$ exactly when I carries the content of R without loss; in particular $\Phi(R, R) = 100$.

F3. Monotonicity. For any interpretive channel Λ operating on the derivative alone, with no access to R and no access to sources correlated with R, the index satisfies $E[\Phi(R, \Lambda(I))] \leq E[\Phi(R, I)]$, the expectation taken over the channel's operation. The requirement binds the average: a single invented output may resemble the reference by luck, and a channel ignorant of the reference loses on average. A channel with side access to R sits outside the Markov chain and may raise the score; that is a return to the source in the sense of Section 24, and it restarts the chain rather than violating the axiom.

F4. Continuity. Φ varies continuously with the strength of interpretive extraction.

Two known evasions test F3, one semantic and one optical, and they are the same evasion. A confident editor who restores a damaged passage from his own model of what it must have said produces text that reads better and informs less: the restoration draws on the editor's prior, mutual information with the source cannot rise, and the gain is plausibility rather than fidelity. This is the Dionysius mechanism of Part Two stated as mathematics; each historian corrected the record towards his own prior, raising its polish and lowering its fidelity. The optical twin is post-selection. A spatial filter raises the visibility of the beam it transmits by discarding light; the surviving ensemble is cleaner and smaller, and the improvement is selection rather than transmission. F3 binds channels acting on the whole ensemble, and the protocols of Part Fourteen fix the optical train so that no measurement gains visibility by narrowing what it looks at. Experiment 1.1 carries the semantic discriminant by construction: at matched compression, the distorted summary must score below the faithful one, however fluently it reads.

F3 is the data processing inequality (14) stated as a design requirement [18]; F4 is Englert's relation (19) stated the same way [24]. The IRI of Part Six is a realisation of this axiom set: the Ω scale supplies F1 and F2 by construction, the optical comparison of Part Seven inherits F3 and F4 from the physics of the channel, and the measurement procedure makes the value of Φ an instrument reading. The axioms define the quantity. The optics realise it.

32. Position among the established measures

Three measures already occupy this territory, and the index owes each a stated relation.

Shannon mutual information I(X;Y) measures the statistical dependence of two ensembles [1]. It satisfies F3; the data processing inequality is its defining behaviour under channels [18]. It is bounded only by the source entropy, it is defined over ensembles of messages, and it assigns no value to a single artefact pair. The IRI keeps mutual information's monotone structure and trades its ensemble character for a per-artefact, bounded, dimension-resolved scale. Mutual information is the theorem's home; the IRI is an instrument built to the theorem's specification.

Quantum fidelity $F(\rho, \sigma)$ measures the closeness of two quantum states on [0, 1], equals 1 exactly on identity, and is non-decreasing under every physical channel [23, ch. 9]. For pure states it reduces to the overlap magnitude $|\langle R|I\rangle|$, which is the quantity a balanced interferometer reads out as fringe visibility. The IRI's core equation (13) is a per-mode intensity comparison: an incoherent, dimension-resolved decomposition of exactly this overlap, while the

full interferometric visibility of $|I\rangle$ against $|R\rangle$ measures the pure-state fidelity itself. Among the established measures, quantum fidelity is the IRI's nearest relative: the IRI is a scaled, mode-resolved fidelity with a declared reference and a declared encoding.

Fisher information $J(\theta)$ measures the sensitivity of a parametrised family of distributions to its parameter, and through the Cramér-Rao inequality its reciprocal lower-bounds the variance of any unbiased estimator [18, section 11.10]. It is local, unbounded, and directional: it prices the detectability of a small change, and it obeys a data processing inequality of its own: processing cannot increase it [25]. Fidelity and Fisher information are two descriptions of one geometry of distinguishability [26]. In that geometry the IRI is an integrated, bounded, conventionally scaled coordinate, and Fisher information is its differential: the sensitivity of the instrument to the smallest distortion it can detect.

The Colour Rendering Index completes the family portrait. Ra is bounded, equal to 100 on the reference illuminant, monotone in colour-shift distance, and continuous. [2]

Measure	Compares	Bounded	Monotone under processing	Per artefact
Mutual information	two ensembles	by source entropy	yes (DPI)	no
Quantum fidelity F	two states	[0, 1]	yes	yes
Fisher information	family against parameter	unbounded	yes	local
CRI (Ra)	source against reference illuminant	≤ 100	yes (in ΔE)	yes
IRI (Ia)	derivative against reference	[0, 100]	required (F3)	yes

CRI satisfies the four axioms for colour. The IRI satisfies them for information.

33. The thermodynamic account

Landauer (1961) proved that erasing one bit dissipates at least $k_{BT} \ln 2$ of heat (k_B: Boltzmann's constant) [27], and Bérut et al. (2012) measured the bound directly in a single-particle trap [28]. The price attaches to erasure. A measurement can in principle be performed reversibly; the unavoidable cost falls due when the instrument resets its registers and overwrites its records. An instrument that runs continuously erases continuously, so a metrological system for information carries a thermodynamic floor set by its erase rate: erased bits per measurement, times measurements per second, times $k_{BT} \ln 2$.

The magnitude deserves honest statement. At room temperature $k_{BT} \ln 2$ is 3×10^{-21} joules; an instrument erasing 10^{15} bits per second pays microwatts against the PSI architecture's kilowatt draw. The engineering budget of Part Five is set by lasers, modulators, and thermal control, and the Landauer floor sits many orders of magnitude below it. The floor's significance is conceptual: information operations are physical operations, the ledger of Section 30 closes in joules, and the archival choice of Part Five follows from it. A record kept costs nothing at the bound; a record erased is paid for. The thermomagnetic tape holds the calibration history as standing physical state because the cheapest bit, thermodynamically, is the bit you keep.

34. The order of claims

The framework asserts a ladder of claims, each rung independently stated and independently testable.

C1, theorem. Interpretive transmission loses information about its source, monotonically, under any processing of the derivative alone. The data processing inequality, Section 24. This rung is proved.

C2, theorem. Physical channels contract distinguishability under the same monotone structure, and optics reads the contraction directly as fringe visibility. The channel formalism of [23], Sections 27 to 30. This rung is proved.

C3, design claim. An index satisfying axioms F1 to F4 can be realised optically, with the IRI of Part Six as the realisation and the instrument of Parts Five and Ten as its embodiment. Phases 1 and 2 of the experimental programme test this rung.

C4, hypothesis. The semantic degradation curve and the optical decoherence curve share a functional form. Experiment 1.2 tests it.

C5, hypothesis. The gauge of the scale, the number of index points one unit of deviation costs, is recoverable from the substrate. Two roles of the constant must be separated, because physics bears on them differently. The grain of the scale is its resolution: the smallest deviation $\delta\Omega_{\min}$ the instrument distinguishes from zero, and therefore the smallest index difference that carries meaning; the grain belongs to the substrate and is tested at stage 2 of Experiment 3.3. The gauge is where the scale's points sit; the gauge is fixed by the anchoring convention of Part Six. Convention and derivation coexist in mature metrology: the International Temperature Scale of 1990 is a conventional scale, and the difference between T90 and thermodynamic temperature is itself a published, re-estimated research quantity [40, 41]. The hypothesis states that a candidate law

$$\kappa_{\text{phys}} = f(\Gamma\tau_m, N) \tag{20}$$

pre-registered before any comparison, reproduces κ_{conv} within combined uncertainty. Experiment 3.3 tests it, and Part Six states the anchoring convention that fixes the scale until it reports.

The rungs are ordered by strength, and the monograph's weight rests on the lower ones. If C5 holds, the physics generates the scale, and the realisation becomes as tight as any in metrology, with the constant derived rather than conventional. If C5 fails, the IRI stands as CRI stands: a defined quantity, a conventional scale, a physical realisation, and an instrument. The programme is ordered so that the poetry is the last thing at risk.

PART TEN: The Encoding Protocol

35. Two measurands

A document puts two distinct questions to a measuring instrument. The first: is this physical object the same object it was yesterday? The second: does this text render the claims of its source faithfully? These are different quantities. They have different references, different failure modes, and different physics. A framework that answers both with one number has confused an object with its content.

Name the quantities. Artefact integrity is the persistence of a physical object through time: the sameness of the paper, the ink, the surface microstructure. Content fidelity is the faithfulness of a derivative's claims to its source's claims: a property of the symbolic layer, indifferent to paper and ink. The four combinations are all physically real:

	Content faithful	Content degraded
Same artefact	The intact original	The original altered in place: the forged entry
Different artefact	A verbatim copy in a new typeface	The distorted derivative

The diagnostic corner is the verbatim copy in a new typeface: a different physical object carrying identical content. An instrument that scores it low has measured the object. An instrument that scores it at ceiling has measured the content. Section 54 makes this corner the discriminating control for the entire encoding protocol: one pair of documents, measured on both arms of the instrument, must return opposite verdicts.

One apparatus, two measurands, and a single control that tells them apart.

36. Artefact integrity: light reads the object

Coherent illumination of a document surface returns a field shaped by everything physical about that surface: glyph geometry, ink distribution, paper fibre microstructure. At the microstructure scale the returned field is a speckle pattern unique to the individual object. Two impressions of the same page from the same press decorrelate at this scale. This uniqueness is established measurement practice: laser-speckle fingerprinting identifies individual paper documents by their surface microstructure (Buchanan et al., 2005 [29]), and scattering microstructure serves as a physical one-way function in authentication systems (Pappu et al., 2002 [30]). Interference recording of an object field against a reference beam preserves that signature as a physical record (Gabor, 1969 [31]).

The consequences fix the measurand's scope. Raw-surface comparison is a same-object measurement. Its reference is the object's own recorded hologram, held on the thermomagnetic tape. Its question is temporal: has this object changed since its state was recorded? Alteration of the surface, whether erasure, overwriting, or substitution, appears as localised decorrelation against the recorded state, and holographic comparison resolves surface change at sub-wavelength scale. Comparison of two different objects returns decorrelation regardless of what either says, which is the correct answer to the authentication question and the reason the raw surface can never serve content comparison. This is Tier 1 measurement in its pure form: zero transduction, no symbolic layer, light interacting with matter.

The scope's edge is equally plain. The raw surface carries no access to the claims the ink spells out. A surface hologram of a false affidavit is exactly as stable as a surface hologram of a true one.

Artefact integrity answers one question completely: this is the same object, unaltered since its record.

37. Content fidelity requires a transducer

Content lives in the symbolic layer, and no optical operation on a raw surface reaches it. To measure content optically, the content must first be projected into a physical representation the optics can compare. That projection is a transduction, and the doctrine of this Part is that it must be declared, specified, and recorded.

Every measuring instrument transduces. A mercury thermometer measures the expansion of mercury; a calibrated relation carries the expansion to temperature. A pyrometer measures spectral radiance and reaches temperature through Planck's law. Metrology's response to this universal fact is declaration: fix the transducer, calibrate it, version it, and record it in the chain of traceability.

Metrology also names the deeper case, the one this framework occupies: quantities defined by their procedure. Rockwell hardness is defined by the indenter, the load, and the protocol of ISO 6508-1; a hardness number carries its scale designation, HRC 40, because the procedure is part of the quantity. pH is defined operationally by the primary-cell procedure of the IUPAC recommendation (Buck et al., 2002 [32]). The International Vocabulary of Metrology recognises quantities whose definition is the measurement procedure itself (JCGM 200:2012 [33]). These are traceable, internationally standardised measurements, and their honesty is structural: the procedure is published, and the number never travels without it.

Content fidelity is a quantity of this class. Definition: content fidelity is the agreement between derivative and source after both are projected through a declared canonicalisation transducer into canonical physical form, measured by optical comparison of the two canonical states. The transducer is part of the measurand. A reported index names its transducer version the way a hardness number names its scale.

38. The canonicalisation transducer

A canonicalisation transducer T is a fixed procedure that maps a document's content to a canonical physical representation. Four properties are mandatory, and together they define the class.

Fixed. T is specified completely before any measurement it serves. Its behaviour on an input is determined by the specification at byte level, with no parameter left to the operator's discretion.

Content-blind in operation. T applies the same deterministic mapping to every input. It has no access to truth values, sources, stakes, or desired outcomes, and no branch of it conditions on what the claims say. It processes a lie and a correction identically.

Versioned. T carries a version identifier. The specification and its implementation are hashed, and the hash is recorded on the thermomagnetic tape with every measurement it serves. Two laboratories running T on the same input produce bit-identical canonical output.

Auditable. Disagreement about T's behaviour is resolved by inspection of the published specification against the recorded inputs and outputs, with no appeal to judgement.

The baseline transducer T1.0 is specified as follows.

Input. The document's character content. Phase 1 restricts inputs to born-digital text or double-keyed transcription with checksum agreement, so that extraction contributes no unmeasured error. Optical character recognition on degraded physical sources is admissible in field use only with its error rate measured and carried as a transducer uncertainty component.

Normalisation. Unicode NFKC, so that ligatures, presentation forms, and compatibility variants fold to canonical characters; soft hyphens removed and line-break hyphenation resolved; quotation and dash glyph variants folded by a declared table; whitespace collapsed. The compatibility folding is a stated convention of T1.0. Two documents are T1.0-equivalent exactly when their normalised character sequences are identical.

Canonical rendering. The normalised sequence is set by a frozen rendering pipeline: one specified typeface at fixed pitch, fixed line grid, fixed page geometry, specified rasterisation. The output raster is hashed and recorded. Typeface, layout, and medium of the input document are annihilated at this step by construction.

Physical realisation. Phase 1 realises the canonical raster as a printed artefact through a frozen print pipeline: one printer model, one paper stock, recorded settings. The print carries the stock's own microstructure, so the content-arm analysis band-limits spatially to the glyph scale, suppressing fibre-scale speckle so the comparison operates on the rendered content rather than the paper. Phase 2 realises the raster directly as structured light through the Spatial Light Modulator already present in the PSI architecture, removing the paper entirely. The transducer is identical in both realisations.

The optical comparison. With source and derivative both in canonical form, the instrument performs an optical cross-correlation of the two canonical states, in the interferometric configuration of Section 54 or equivalently in a matched-filter or joint-transform correlator geometry. Optical correlation of this kind is a demonstrated primitive of six decades' standing (Vander Lugt, 1964 [34]; Weaver and Goodman, 1966 [35]). The contribution here is the metrological wrapper: the declared transducer, the recorded chain, the reference doctrine. The optical primitive itself is classical.

What T1.0 measures is declared without decoration: divergence between canonical character sequences, detected and localised optically. Zero divergence maps to the ceiling score by construction; this is the verbatim-invariance property, and it is the property the discriminating control of Experiment 1.1 tests. Any divergence produces a deviation in the canonical states, which the correlation detects and the spatially resolved interference localises. The magnitude of optical deviation tracks the extent of textual change. The severity of a change is a semantic quantity, assigned in the dimensional layer (D1 to D14), and this monograph makes no claim that the physical layer weighs it: one inverted negation can matter more than a hundred synonym substitutions, and under T1.0 the instrument reports both as localised divergence while the dimensional layer assigns the weight.

Paraphrase is declared the same way. T1.0 registers a faithful paraphrase as divergence, so the quantity measured under T1.0 is textual fidelity: the strictest and least interpretive member of the content-fidelity family. Transducers that canonicalise claims rather than characters, a T2.x family, would move the measurand towards semantic fidelity at a declared cost: a heavier specification, more contestable, harder to hold content-blind. The doctrine admits them on one condition: they satisfy the same four properties. Whether a claim-level canonicalisation can remain content-blind is an open question (Open Questions).

The transducer is declared, versioned, and part of the result; the number never travels without it.

39. The recursion argument, restated

The objection this Part exists to answer runs: any measurement of content requires reading content; reading is interpretation; therefore interpretation stands at the base of the instrument, and the index measures the encoder's opinion. Interpretation all the way down.

The answer is the oldest move in metrology: replace an open-ended judgement with a frozen procedure. The regress requires each layer to exercise discretion. A transducer has none. T is fixed before the content arrives; it conditions on nothing the content says; it produces bit-identical output in any laboratory; every decision it makes is inspectable against a published specification. An editorial judgement behaves in the opposite way on every axis: it adapts to the content, weighs the stakes, varies between editors, and resists audit. The difference between a transducer and an interpreter is the difference between a ruler and an opinion about length.

The honest count follows. Artefact integrity carries zero transduction. Content fidelity carries exactly one, the same one, on every tier, declared and recorded on the tape.

One choice remains prior to measurement, and the doctrine states it rather than buries it: the choice of T is consequential. Different transducers define different members of the content-fidelity family, exactly as Rockwell and Vickers define different hardness quantities, and metrology's discipline for this is naming. The IRI adopts it: every reported index is indexed by transducer version. The choice of canonical form is a convention; the measurement under a declared convention is physics.

The regress ends where the procedure freezes. Light does not have an opinion. That is the property that makes it the right substrate.

40. The reference hierarchy under two measurands

The tier hierarchy grades the provenance of the reference. The transduction count is a property of the measurand, and it is constant within each measurand. For artefact integrity, only Tier 1 exists: the reference is the object's own recorded hologram, and the transduction count is zero. For content fidelity, every tier carries exactly one transduction, the canonicalisation transducer of Section 38, identical across tiers. The tiers differ in where the reference content comes from, and in the uncertainty the reference carries.

Physical metrology arranges its references in a calibration hierarchy: a primary standard realises the definition of the quantity; a secondary standard is calibrated against the primary; a working standard serves the bench, the standard used routinely to calibrate or verify measuring instruments (JCGM 200, the International Vocabulary of Metrology [33]). Each step in the hierarchy is a calibration, each calibration contributes uncertainty, and every certificate states the addition. The Guide to the Expression of Uncertainty in Measurement supplies the machinery: every imperfect input to a measurement contributes a quantified uncertainty component, the components combine by stated rules, and the result is reported with its combined uncertainty [36]. The three reference tiers form this structure under different names, and v2 names it properly. A tier is defined by two properties: how its reference content is realised, and the reference uncertainty u(R) the realisation carries.

Tier 1: the primary standard. The content of a physical or born-digital primary source in hand, canonicalised under T; for artefact integrity, the object's own recorded hologram. $u(R_1)$ is evaluated by repeated realisation (Type A evaluation, GUM 4.2): realise the same source's

reference state repeatedly under the standard protocol, compare each state against the pooled state, and take the dispersion, expressed in Ω. A source compared against its own reference must score at the ceiling, and the spread of that self-comparison is the realisation noise floor. The transducer contributes its own component, u_T, characterised by the re-rendering control of Experiment 1.1 and fixed per transducer version. Tier 1 minimises u(R). No realisation abolishes it.

Metrology has never certified a reference at zero uncertainty. The ensemble of national prototype kilograms, the most carefully guarded artefacts in the history of measurement, drifted relative to the international prototype by an average of some fifty micrograms over a century; the drift was measured, published, and eventually helped retire the artefact in favour of a definition fixed to a constant of nature [37]. A stated uncertainty is the mark of a serious reference.

Tier 2: the working standard. Content triangulated from independent live sources and projected via SLM under the same T. The construction protocol stands: independent source collection across multiple platforms (OSINT); propagation analysis identifying which signals are independent and which are correlated echoes (QPN); weighting by source independence, temporal priority, and internal consistency; calibration against known ground truths from the VNS governor framework. Each stage now carries a named uncertainty component, defined in Section 41, and $u(R_2)$ is evaluated from four such components: constituent dispersion corrected for effective independence; weighting sensitivity; calibration residual; and a common-origin allowance. The interpreter enters the system here, and here it is measured. Tier 2 admits interpretation openly and prices it in the unit of the measurement.

Tier 3: a working standard with a partially documented chain. The reference is built from surviving historical sources, each itself scored by the instrument. The constituents of a historical record share transmission lineage. Livy and Dionysius drew on a common annalistic tradition, so their agreement is partly echo; the founding example of Part Two returns here as a quantitative constraint. Where independence between constituents cannot be established, the GUM's conservative treatment applies: correlation is set to its bounding value, constituent uncertainties combine linearly, and $u(R_3)$ is never smaller than the largest constituent uncertainty [36]. The v1 rule, that the reference inherits the lowest score of its constituents, survives inside this treatment as the dominance bound. Where textual scholarship establishes independent lines of transmission, partial quadrature combination is licensed for the independent branches; the stemma codicum is propagation analysis practised on manuscripts, and the stemma itself joins the budget on the tape.

The hierarchy orders references by the uncertainty they can honestly state. Tier 1 minimises u(R). Tier 2 quantifies it. Tier 3 bounds it from below. The tier names the reference's provenance; the transducer version names the measurand; the tape records both, beside the budget that both feed.

41. The uncertainty budget

Every IRI measurement is reported with its combined standard uncertainty, evaluated under JCGM 100 [36]. The budget has three components:

$$u_c^2 = u_{\text{inst}}^2 + u_T^2 + u^2(R) \tag{21}$$

where u_{inst} is the instrument contribution, evaluated by Type A methods from the repeatability and reproducibility trials of Experiment 2.3; u_T is the transducer contribution, evaluated from the re-rendering control of Experiment 1.1 and fixed per transducer version; and u(R) is the reference contribution, evaluated per tier as defined in Section 40. Equation (21) is the GUM's law of propagation of uncertainty for independent components; where components correlate, covariance terms enter per GUM 5.2. The VIM defines an uncertainty budget as the statement of a measurement uncertainty, of its components, and of their calculation and combination [33]; the tape record of Section 14 is that statement in physically auditable form.

For a Tier 2 reference the budget resolves further:

$$u^2(R_2) = s^2(R)/n_{\text{eff}} + u_w^2 + u_{\text{cal}}^2 + u_{\text{com}}^2 \tag{22}$$

Four components. Each answers a named objection.

s(R) is the dispersion of the constituent sources about the synthesised reference, measured by the instrument itself: each constituent is separately rendered and projected, constituent pairs are interfered, and the spread is read out in Ω (Type A). Disagreement among sources becomes a measured input, and the objection that triangulation hides disagreement is answered by a number.

n_{eff} is the effective number of independent constituents. Apparent corroboration inflates when nominally independent sources echo one another. The QPN propagation analysis estimates n_{eff} from the correlation matrix of the constituents' distortion fingerprints (Section 23): compute the matrix's eigenvalues η_i and take the participation ratio,

$$n_{\text{eff}} = (\Sigma_i \eta_i)^2 / \Sigma_i \eta_i^2 \tag{23}$$

Fully independent constituents give n_{eff} equal to their count; perfect echoes give $n_{eff} = 1$. Two outlets with matching distortion fingerprints are one source wearing two mastheads, and equation (23) counts them once. The participation ratio is the working estimator; where the full correlation matrix is retained on the tape, the exact GUM 5.2 combination supersedes it.

u_w is the weighting sensitivity. The construction protocol assigns weights for independence, temporal priority, and internal consistency, and the assignment is interpretive discretion. The discretion is quantified as a Type B component: sweep each weight across its defensible range, record the induced shift in the synthesised reference, and treat the half-width a of that shift as a rectangular distribution, $u_w = a/\sqrt{3}$ (GUM 4.3.7). The interpreter's freedom is a parameter range, and the sweep converts the freedom into a number.

u_{cal} is the calibration residual: the standard deviation of constructed-reference error on held-out ground-truth cases from the governor-framework calibration set (Type A).

u_{com} is the common-origin allowance. Constituents that pass the fingerprint independence test can still share an evidentiary origin: one wire service, one official briefing, one press release rewritten five ways. Shared origin is shared bias, and no amount of averaging reduces it. The provenance machinery (D1) classifies constituents by documented origin, and u_{com} is evaluated Type B from the spread between distinct origin classes. Where the origin taxonomy cannot be established, u_{com} is unbounded, the reference is unfit for purpose, and the instrument declines the measurement and states why.

The budget is classical measurement statistics throughout. No component of u_c inherits a quantum limit, and the monograph claims none: the decoherence framing of Part Nine concerns the measurand; the budget concerns the measurement of it.

The reporting convention. Every certified IRI result is reported as

$$Ia \pm U \qquad U = k \times u_c, k = 2 \tag{24}$$

with coverage factor k = 2 giving approximately 95% coverage under the GUM's near-normality conditions [36], together with the reference tier, the reference identifier, the transducer version, and the tape address of the full budget. A worked form: Ia = 62 ± 4 Ω (k = 2; Tier 2 reference R-2026-041; transducer T1.2; budget on tape). The coverage factor is always stated so that the standard uncertainty can be recovered downstream.

Two measurements are distinguished only when their separation clears their combined uncertainties:

$$\left|Ia^{(1)} - Ia^{(2)}\right| > k\sqrt{u_{c,1}^2 + u_{c,2}^2} \tag{25}$$

Differences inside this bound are noise. The monograph's own claims are subject to the same arithmetic: a cascade layer, a corpus comparison, a governor trigger is asserted only where inequality (25) holds.

The budget propagates through the framework's own laws. For the multiplicative cascade of Section 25, relative uncertainties combine in quadrature across layers: $(u_M/M)^2 = \Sigma_j\ (u_j/M_j)^2$. The instrument that measures degradation along a chain also states how confidently it has done so, link by link.

Fitness for purpose closes the loop between budget and application. The VIM defines target uncertainty as an upper limit on measurement uncertainty set by the intended use of the result [33]. The working rule: a reference is fit for a measurement when U is at most half the smallest fidelity separation the measurement must resolve. To resolve a single professional interpretive layer under the anchoring convention of Section 17, a 10 Ω separation at the top of the scale, the budget must deliver $U \leq 5\ \Omega$. A reference that misses the target for one application is refused for that application and remains usable where the target is looser.

Section 39 froze the transducer and ended the regress on the measurand's side. Reference construction posed the residual objection: on Tier 2 the interpreter returns through the selection and weighting of constituents. The budget is the answer, in metrology's own terms. An unstated interpreter is a bias. A budgeted interpreter is an uncertainty component with a stated magnitude, a stated evaluation method, and an audit trail on the tape.

Converting hidden bias into stated uncertainty is the founding operation of modern metrology, and the IRI performs it on the interpreter itself.

42. Self-auditing references

Constructed references can themselves be measured, and v2 assigns that property a job: the self-audit is the procedure that evaluates u(R).

The Tier 2 reference has measurable provenance (D1), internal coherence (D3), and source plurality (D4), and the same instrument measures all three. Each audit output feeds a named budget component. The plurality machinery, pairwise interference of constituent projections, yields the fingerprint correlation matrix from which n_{eff} is computed. The coherence machinery yields the constituent dispersion s(R). The provenance machinery, the documented origin chain on the tape, fixes the origin taxonomy that bounds u_{com}. The audit returns numbers, the numbers populate equation (22), and equation (22) qualifies the certificate. A reference that

fails its own audit is withdrawn from service exactly as a drifting gauge block is withdrawn from a calibration bench.

The comparison with prior art sharpens. The Admiralty Code grades source reliability on an ordinal human scale, A to F and 1 to 6, with no propagation rule; two B2 sources compound into an unquantified judgement. The IRI budget is interval-valued and propagates: constituents with stated uncertainties combine by stated arithmetic into a reference with a stated uncertainty, which combines with the instrument's noise into a measurement with a stated uncertainty. Doubt acquires a chain of custody that runs unbroken from constituent to certificate.

This is the VIM's definition of metrological traceability applied to information: a result related to a reference through a documented unbroken chain of calibrations, each contributing to the measurement uncertainty [33]. The metrology is recursive in the right direction: each layer of the chain is auditable by the same instrument, down to the holographic bedrock of Tier 1, and every audit returns a quantity the budget consumes.

A reference that states and measures its own uncertainty is a working standard in the full sense of JCGM 200. The framework's most exposed component becomes its most conventional: Tier 2 is what every calibration laboratory on earth already does, performed in light.

PART ELEVEN: The Unit

43. The render

The render (Ω) is the unit of information rendering fidelity: the OAM mode power ratio of a single information dimension measured against a defined reference state, under a declared canonicalisation transducer version.

100 Ω: perfect fidelity.

0 Ω: total information loss.

Ia: mean of the per-dimension I_i values, each individually passed through the soft-floor transform of equation (9) before averaging. Ia is reported with its transducer version, as Ia (T1.0), the way a hardness number carries its scale. Artefact integrity is reported separately as V_A and never averages into Ia.

44. Derived measures

Ω-spectrum: distortion fingerprint. Diagnostic of distortion type.

Ω-cascade: cumulative degradation across layers. What VB measured.

Ω-rate: degradation velocity. What VNS proposed regulating.

ΔΩ: deviation on a single dimension.

45. The metrological chain

For artefact integrity:

Object → recorded hologram (tape) → later re-illumination → correlation against record → V_A

The artefact chain carries zero transduction; its uncertainty budget comprises the instrument term u_{inst} and the realisation noise $u(R_1)$ from the recorded-hologram reference (Section 40), with $u_T = 0$.

For content fidelity, Tier 1:

Source content → T (version and input hash on tape) → $|R_c\rangle$ → thermomagnetic record → OAM measurement → $\Delta\Omega_i$ → Ia ± U (T version)

with the derivative passing through the identical chain to $|I_c\rangle$. Dominant budget components: u_{inst} and u_T.

For content fidelity, Tier 2:

Constituent collection (OSINT) → independence analysis (QPN: n_{eff}) → weighted synthesis → governor-framework calibration → T (version and input hash on tape) → SLM projection → $|R_c\rangle$ → thermomagnetic record (states and budget) → OAM measurement → $\Delta\Omega_i$ → Ia ± U (T version)

Dominant budget components: the four terms of equation (22).

Every link is physical or budgeted, and most are both. The tape carries the states, the transducer version, the input hashes, and, for Tier 2, the full construction record and its budget as one record; a certificate that cannot be reconstructed from the tape is void. The chain satisfies the VIM's definition of metrological traceability: an unbroken, documented sequence of calibrations, each contributing stated uncertainty, connecting the reported Ia to the realisation of its reference [33].

PART TWELVE: Unification

46. One problem, three papers

Velocity Bias identified the symptom: information degrades through retransmission, and this degradation contaminates AI systems. Vital Network Science proposed the treatment: regulate the transmission rate, and leave the content alone. Photonic Substrate Intelligence built the instrument: a physical substrate where light performs computation without interpretive overhead.

Each investigation was conducted independently. Their coherence was unplanned. It became visible only in retrospect, when the theory presented in this monograph revealed them as three perspectives on the same underlying absence: the absence of physical information metrology.

47. The correspondence

Concept	VB (2025)	VNS (2025)	IRI-PSI (2026)
Quality metric	$\psi(c)$	$\psi(t)$	Ia (Ω)
Degradation	$\beta\varphi > 1$	$d\psi/dt < 0$	Ω-cascade
Velocity	$\dot{\varphi}(c)$	$d\varphi/dt$	Ω-rate
Intervention	Vitalic curation	Governor G	Physical measurement
Basis	Statistical	Control-theoretic	Physical (decoherence)
What it is	The symptom	The treatment	The diagnosis

The relationship is convergent. Three paths to the same centre.

PART THIRTEEN: Implications

48. For epistemology
Information degradation through interpretation obeys the mathematics of decoherence, and light makes that mathematics measurable. Epistemology acquires an instrument.

49. For AI safety
Velocity bias is cascade degradation at speed. The IRI provides a physical measurement of training data fidelity. AI systems can be evaluated by the Ω-spectrum of their training corpus. VB's feedback loop prediction, that AI-generated content compounds the distortion, becomes measurable and therefore preventable.

50. For information governance
VNS proposed changing the optimisation target from φ to ψ. The IRI makes this calibrated rather than arbitrary. The governor G triggers on Ω-rate, a physically measurable quantity. The treatment becomes prescribable because the diagnosis is now physical.

51. For computation
PSI is computation with built-in metrology. Every inference carries its fidelity measurement. This is the compiler principle: the measurement is in the machine code, and the instrument of Parts Five and Ten is the first that can measure the quality of what it computes.

52. For society
People will still lie to each other. The public information infrastructure acquires a weights and measures act. Distortion becomes detectable, provable, and costly.

PART FOURTEEN: Experimental Programme

53. Programme structure

The theoretical framework is complete. What follows is the experimental programme required to validate it. The programme is structured in three phases, ordered by dependency: each phase's results determine whether the subsequent phase is warranted. Experiments within each phase are ordered by criticality; the experiment most likely to falsify the framework is performed first, so that resources are not expended on downstream validation if the foundation does not hold.

All experiments are designed to be reproducible by any laboratory with standard quantum optics and photonics equipment. No proprietary hardware is required for Phases 1 or 2. Phase 3 requires components of the PSI architecture. Detailed protocols, sample sizes, statistical methods, and decision criteria are specified for each experiment to enable independent replication.

54. Phase 1: Foundation

Phase 1 tests whether the core physical claims hold. These are the experiments that determine whether the framework has a future. If any Phase 1 experiment returns a negative result, the framework must be revised or abandoned before proceeding.

Experiment 1.1: Measurand separation and encoding validation

Question. Content arm: after canonicalisation under T1.0, does optical comparison produce a signal that is invariant to form, sensitive to minimal content alteration, and able to separate faithful from distorted derivatives at matched compression? Artefact arm: does raw-surface holographic comparison detect physical alteration of an object against its own recorded state, and does it identify a distinct physical object as distinct?

This remains the most critical experiment in the programme. It tests the measurand separation itself: whether the instrument's content channel responds to content and its artefact channel responds to the object, each blind to the other's variable.

Method. Select 50 source documents of diverse types: official government records, peer-reviewed papers, legal filings, journalistic reports, and historical texts; born-digital or double-keyed per the T1.0 input rule. For each source, construct a six-item derivative panel in which form, length, and content vary independently:

(a) Verbatim copy in the source's native form.
(b) Verbatim copy re-rendered in a different typeface, layout, and medium: reflowed single column, different fount, different page size. This is the discriminating control.
(c) Minimal alteration: exactly one load-bearing token changed, a date, a figure, a name, or a negation inserted or removed, in an otherwise verbatim text in native form. Edit positions logged and sealed until analysis.
(d) Faithful summary at 25% compression, produced by a professional analyst instructed for maximum fidelity.
(e) Distorted summary at 25% compression, produced by the same analyst pool with three specified factual distortions seeded per document, logged and sealed.
(f) Heavily editorialised full-length rewrite, length within 10% of the source.

All six items pass through T1.0 for the content arm. For 20 of the 50 sources, fabricate an additional physical panel for the artefact arm:

(g) The source object itself, re-imaged after re-mounting and handling: the self-match floor.

(h) The object after controlled physical alteration: chemical erasure of one token, overwriting, or page substitution, matched where possible to the content edit of item (c).

(i) A best-effort facsimile: the same content reprinted on the same stock in the same typeface.

Apparatus. Coherent laser source ($\lambda = 632.8$ nm HeNe or 532 nm frequency-doubled Nd:YAG). Beam expander and spatial filter. Document mount with precision positioning. High-resolution CCD or CMOS camera (minimum 4096×4096 pixels). Optical bench with vibration isolation. Beam splitter for Mach-Zehnder interferometric comparison. Addition over the prior list: the frozen print pipeline of Section 38 (one printer model, one stock, recorded settings; under £1,000).

Procedure. Content arm: (1) Render the source and each derivative through T1.0; hash and record every canonical raster. (2) Realise the canonical rasters physically through the frozen print pipeline. (3) Interfere source-canonical against derivative-canonical in the Mach-Zehnder configuration; capture the spatially resolved interference; compute global visibility V_c and the deviation map, band-limited to the glyph scale per Section 38. (4) Image item (a) ten times with re-mounting to establish the repeatability distribution; fix the equivalence margin δ in advance as the 95th percentile of $|V_c(a) - V_c(a')|$ across repeats. Artefact arm: (5) Record each physical object's raw-surface hologram; re-illuminate and correlate for items (g), (h), and (i), and for the physical print of item (b) against the source's recorded hologram. Analysis is blinded: the analyst computing detection and separation statistics receives patterns without condition labels, and the sealed edit logs are opened only at reconciliation.

Measurements. Content arm: V_c per pair; deviation maps; localisation accuracy against the sealed edit logs for item (c); paired separation statistics for items (d) against (e). Artefact arm: correlation against the recorded state; detection statistics for item (h); rejection rate for item (i).

Predicted outcomes and success criteria. Each prediction is labelled with what its failure kills.

P1, form invariance: the discriminating control. $V_c(b)$ is statistically equivalent to $V_c(a)$ within margin δ (two one-sided tests on the paired differences, $\alpha = 0.05$), and the physical print of item (b) falls below the artefact match threshold on the raw arm. One document pair, two measurands, opposite verdicts: this single result displays the entire separation. The control's teeth are at the specification's seams: ligatures, soft hyphens, and glyph variants are exactly where form can leak through an imperfect normalisation, and any residue appears here. Failure: form leaks through the transducer and the encoding reads form; T1.0 is falsified as a canonicalisation and enters the failure response below.

P2, minimal-alteration sensitivity. For item (c): a localised deviation exceeding three standard deviations of the (a)-repeatability noise map at the logged edit site in at least 90% of the 50 pairs, with a false-alarm rate of at most 5% on items (a) and (b). Failure: the encoding's sensitivity floor sits above the single-token scale; the floor is then measured and published, and content-fidelity claims are narrowed to divergence above it.

P3, content separation at matched compression. $V_c(d) > V_c(e)$, paired across the 50 sources, effect size $d_z \geq 0.5$ at $p < 0.01$ (paired test; Wilcoxon signed-rank if normality fails). With $n = 50$ the design's power is approximately 0.8 for the smallest effect the criterion accepts. Length is held constant within each (d)-(e) pair, so a positive result is attributable to content. This criterion supersedes the earlier five-level monotonicity prediction, in which degradation levels co-varied with length and format and a monotonic trend was therefore obtainable from form

sensitivity alone. Failure: optical divergence under canonicalisation fails to track seeded factual distortion at matched length; the semantic-tracking claim is falsified even with form controlled; the instrument stands as a divergence detector and the dimensional-mapping programme of Experiment 1.4 is re-scoped before any further phase.

P4, descriptive ordering. Predicted ordering of content-arm scores: (a) and (b) at ceiling; (c) at ceiling globally with a localised flagged deviation; (d) intermediate, reflecting omission; (e) below (d); (f) below (a). The ordering of (e) against (f) is reported as measured: the framework ranks distortion against omission by measurement alone. P1 to P3 are pass/fail; P4 is reported.

P5, artefact arm. Self-match (g) above the match threshold on every re-mount. Physical alteration (h) detected at sensitivity $\geq 95\%$ and specificity $\geq 95\%$ under blinded scoring. Facsimile (i) rejected as a distinct object in $\geq 99\%$ of trials, consistent with the established uniqueness of paper-surface microstructure (Buchanan et al., 2005 [29]). Failure of the facsimile rejection contradicts the published fingerprinting result and triggers an apparatus review before any interpretive conclusion is drawn.

Success criterion. P1 and P3 must both pass for the content-fidelity programme to proceed. P5 must pass for the artefact-integrity applications to proceed. The two verdicts are independent: a failure in one arm leaves the other standing. This is the redesign's structural gain: one brittle composite claim becomes two independently falsifiable claims.

Failure response. If P1 fails: audit the transducer for form leakage (fount metrics, normalisation seams, rasterisation artefacts, print-pipeline variance), repair, increment the version, and re-run the control. If no transducer version achieves form invariance, the optical content-fidelity claim is withdrawn; the framework retains artefact integrity as its physically grounded measurement and content fidelity reverts to a computational model. If P3 fails after P1 passes: the canonical encoding distinguishes verbatim from non-verbatim but fails to grade degradation, and the index's content claims contract to divergence detection and localisation.

Resources. Standard optics laboratory; equipment £15,000 to £25,000 if not already available, plus the print pipeline under £1,000. Personnel: 1 experimental physicist, 1 research assistant; analyst fees for the (d) and (e) panels within the envelope. Duration: 8 to 12 weeks. Estimated total cost: £30,000 to £50,000.

Decision gate. Positive P1 and P3 are required before any content-dependent experiment proceeds: this is the go/no-go gate for the content programme. Positive P5 gates the artefact-integrity applications. The programme halts only if both arms fail.

Experiment 1.2: Graduated observation and the Englert mapping
Question: Does Englert's duality relation $D^2 + V^2 \leq 1$ map quantitatively onto information fidelity degradation, such that partial interpretive extraction produces partial fidelity loss on a continuous scale?

Method. Construct a which-path interferometer with continuously variable measurement strength (e.g., adjustable polarisation-based path marking). At each measurement strength setting, record both the distinguishability D (how much path information is extracted) and the fringe visibility V. Simultaneously, for the same source documents used in Experiment 1.1, produce derivatives at continuously variable degradation levels (e.g., increasing summarisation compression ratios from 100% to 10% of original length). At each compression level, measure the holographic fringe visibility from Experiment 1.1. All document derivatives in the compression series are measured on the content arm under T1.0, so the compression variable operates on canonical renderings rather than on native layouts.

Measurements. Plot $D^2 + V^2$ for the quantum optical system. Plot analogous fidelity-extraction curve for the document degradation series. Compute the functional form of both curves. Test whether the document degradation curve follows the same inequality constraint as the quantum system.

Predicted outcome. Both curves obey $D^2 + V^2 \leq 1$ or a closely analogous constraint. The document fidelity curve is continuous and monotonically decreasing, with functional form statistically indistinguishable from the quantum optical curve (Kolmogorov-Smirnov test, $p > 0.05$).

Success criterion. Document degradation curve must be continuous (not binary), monotonically decreasing, and bounded by a constraint of the form $F^2 + V^2 \leq 1$ where F is the fidelity extraction measure. If the curve is discontinuous, non-monotonic, or unbounded, the Englert mapping is not supported.

Resources. Quantum optics laboratory with variable-strength which-path interferometer. Duration: 6 to 8 weeks. Estimated cost: £20,000 to £35,000.

Experiment 1.3: Cascade degradation measurement
Question: Does information fidelity degrade multiplicatively across interpretive layers, as predicted by the cascade equation $M_i(\text{total}) = \prod_j M_{ij}$?

Method. Select 20 source documents. For each, construct a chain of $L = 10$ interpretive layers: source → analyst summary → editorial rewrite → translation → back-translation → journalistic report → social media summary → second-hand share → political reframing → public memory recall. Each layer is performed by a different human participant (total: 200 participants, recruited and compensated). At each layer, holographically image the output document using the protocol from Experiment 1.1. Compute fringe visibility V at each layer relative to the original source. Each layer's output is canonicalised under T1.0 before imaging; without this step the cascade curve would confound content drift with form drift, since a social-media summary arrives in a different form by construction.

Measurements. V at each layer for each document chain. Fit the decay curve to both multiplicative model ($V_L = V_0 \times m\hat{}L$) and additive model ($V_L = V_0 - a \times L$). Compare fit quality (adjusted R^2, AIC, BIC). Compute per-layer fidelity coefficient m across document types. Test whether m is consistent across document categories.

Predicted outcome. Multiplicative model provides significantly better fit than additive model ($\Delta\text{AIC} > 10$). Per-layer fidelity coefficient m falls in range 0.85 to 0.95. Coefficient m varies by document type (predicted: legal/technical documents retain higher per-layer fidelity than political/emotional content).

Success criterion. Multiplicative model must provide superior fit. If additive model fits equally well or better, the cascade equation is not supported and the compounding-degradation claim must be revised.

Resources. 200 participants at £50 each (£10,000). Optical equipment from Experiment 1.1. Personnel: 1 experimental physicist, 1 research coordinator, 2 research assistants. Duration: 12 to 16 weeks. Estimated total cost: £40,000 to £60,000.

Experiment 1.4: Dimensional structure analysis
Question: How many independent dimensions of information distortion exist in the holographic encoding, and do they correspond to the proposed D1-D14 test dimensions?

Method. Using the 300 source-derivative pairs from Experiment 1.1 (50 sources × 6 derivatives), plus the 200 cascade intermediates from Experiment 1.3, extract the full diffraction pattern feature vectors (spatial frequency components, phase distributions, intensity profiles) from content-arm canonical measurements only; artefact-arm data are excluded from the dimensional analysis. Independently, have trained human assessors score each derivative on the 14 proposed IRI dimensions (D1-D14) using a standardised rubric. Perform principal component analysis (PCA) on the diffraction pattern feature space. Compare the PCA eigenvectors against the human-assessed dimension scores using canonical correlation analysis (CCA).

Measurements. Number of PCA eigenvalues above noise floor (this determines the true dimensionality of information distortion space). Canonical correlations between PCA components and human-assessed D1-D14 scores. Percentage of variance in human assessments explained by the photonic PCA components.

Predicted outcome. 6 to 12 eigenvalues above noise floor (true dimensionality of the distortion space). Significant canonical correlations ($r > 0.4$) between at least 6 PCA components and corresponding human-assessed dimensions. At least 60% of variance in human assessments explained by photonic PCA components.

Success criterion. At least 4 PCA components must show significant canonical correlation with human-assessed dimensions. If the diffraction patterns contain no systematic dimensional structure that maps to human-assessable information fidelity, the claim that the photonic encoding captures IRI dimensions is falsified.

Note. This experiment determines the true basis set for OAM mode assignment. If the empirical dimensionality is N, then N OAM modes are required, regardless of whether N matches the proposed 8 or 14. The physics determines the dimensions; the theory adapts.

Resources. Data from Experiments 1.1 and 1.3. 10 trained human assessors at £500 each (£5,000). Statistical analysis software. Duration: 8 weeks (can run concurrently with Experiment 1.3). Estimated cost: £15,000 to £20,000.

55. Phase 1 summary

Exp.	Tests	Duration	Cost
1.1	Measurand separation: content and artefact arms	8-12 weeks	£30k-£50k
1.2	Englert mapping to information	6-8 weeks	£20k-£35k
1.3	Cascade multiplicative law	12-16 weeks	£40k-£60k
1.4	Dimensional structure and basis set	8 weeks (concurrent)	£15k-£20k

Phase 1 total: 16 to 24 weeks elapsed (with concurrency). £105,000 to £165,000. Requires one equipped optics laboratory and a team of 4 to 5 researchers.

Phase 1 decision gate: Experiment 1.1 gates by arm. P1 and P3 (content) are required for every content-dependent experiment; P5 (artefact) is required for the artefact-integrity applications. If the content arm fails, the programme continues on artefact integrity alone and the content framework reverts per the failure response. If Experiment 1.3 fails, the cascade model is revised and retested before proceeding. Results reviewed by an independent panel of three: one quantum optics physicist, one information scientist, one metrologist. Unanimous positive recommendation required to proceed to Phase 2.

56. Phase 2: Instrument

Phase 2 builds and tests the first prototype measurement instrument. It requires positive Phase 1 results and access to photonic computing components. Phase 2 experiments are ordered by the PSI critical-path dependencies identified in the original architecture paper.

Experiment 2.1: OAM mode fidelity encoding

Question: Can the empirically determined dimensions of information distortion (from Experiment 1.4) be encoded into separate OAM modes and independently measured?

Method. Using the PCA basis set from Experiment 1.4, assign each principal component to a distinct OAM mode ($\ell = 1, 2, \ldots, N$ where N is the empirical dimensionality). Encode the holographic diffraction pattern of a test document onto a beam using a spatial light modulator configured as an OAM mode encoder. At the detector, use an OAM mode sorter to separate the individual modes. Measure power in each mode. Compare against the reference state power distribution. The SLM realisation of T replaces the print pipeline of Phase 1; the transducer specification, version, and hashes are unchanged.

Measurements. Mode fidelity M_i per channel. Cross-talk matrix C_{ij} between channels. IRI score Ia computed from mode fidelity measurements. Correlation between photonically measured Ia and human-assessed IRI scores from Phase 1.

Predicted outcome. Cross-talk $C_{ij} < 0.05$ for $i \neq j$ (less than 5% power leakage between channels). Photonically measured Ia correlates with human-assessed IRI at $r > 0.7$.

Success criterion. Cross-channel isolation must be sufficient to distinguish independent dimensions. Correlation between physical and human-assessed IRI must be $r > 0.6$. Below this, the OAM encoding does not capture the dimensional structure with sufficient fidelity.

Resources. OAM mode encoder (SLM, commercially available). OAM mode sorter (custom optics or commercial unit). Detector array. Duration: 12 to 16 weeks. Estimated cost: £80,000 to £120,000 including equipment.

Experiment 2.2: Thermomagnetic calibration record

Question: Can the measurement process and reference state be recorded on a thermomagnetic substrate with sufficient fidelity to provide an auditable calibration chain?

Method. Write the reference state $|R\rangle$ and measurement parameters onto commercial magneto-optical disc media using a modified MO drive. Read back after 1 hour, 24 hours, 1 week, 1 month. Compare readback state against original write state. Intentionally alter a portion of the recorded state and test whether the alteration is detectable via Kerr rotation measurement.

Predicted outcome. Readback fidelity $> 99.9\%$ at all time intervals. Intentional alteration detected with $> 99\%$ reliability.

Resources. Commercial MO drive and media. MOKE measurement apparatus. Duration: 8 weeks. Estimated cost: £15,000 to £25,000.

Experiment 2.3: End-to-end prototype

Question: Can the complete measurement chain, from document illumination through holographic encoding, OAM mode separation, mode fidelity measurement, to IRI computation, be performed as a single physical process producing a reliable, repeatable score?

Method. Assemble the components from Experiments 1.1, 2.1, and 2.2 into an integrated optical bench configuration. Process 100 document pairs (source plus derivative) through the complete chain. Compute IRI scores. Independently have human assessors score the same

pairs. Test repeatability (same document measured 10 times), reproducibility (same document measured on two independently aligned systems), and validity (correlation with human assessment). Repeatability and reproducibility targets apply to the canonical chain end-to-end, from input hash to Ia, with the transducer version recorded per measurement.

Measurements. Intraclass correlation coefficient (ICC) for repeatability (target: ICC $>$ 0.90). ICC for reproducibility (target: ICC $>$ 0.80). Pearson correlation with human assessment (target: r $>$ 0.70). Mean absolute deviation between repeated measurements (target: $<$ 3 points on 0-100 scale).

Success criterion. ICC repeatability $>$ 0.85. ICC reproducibility $>$ 0.75. Correlation with human assessment $>$ 0.65. If these thresholds are not met, the prototype does not constitute a reliable measurement instrument and requires redesign before Phase 3.

Resources. All Phase 1 and Phase 2 equipment. 20 human assessors at £500 each (£10,000). Duration: 12 to 16 weeks. Estimated cost: £50,000 to £80,000 (excluding prior equipment).

57. Phase 2 summary

Exp.	Tests	Duration	Cost
2.1	OAM mode fidelity encoding	12-16 weeks	£80k-£120k
2.2	Thermomagnetic calibration record	8 weeks	£15k-£25k
2.3	End-to-end prototype	12-16 weeks	£50k-£80k

Phase 2 total: 24 to 32 weeks elapsed (with concurrency). £145,000 to £225,000. Requires photonic laboratory with OAM encoding and mode sorting capability.

Phase 2 decision gate: Experiment 2.3 must demonstrate repeatable, reproducible measurement. Results reviewed by the same independent panel plus one additional metrologist. Majority positive recommendation required to proceed.

58. Phase 3: Validation at scale

Phase 3 applies the validated instrument to real-world information products at scale, and connects the IRI measurement to the velocity bias and vital network science frameworks.

Experiment 3.1: Velocity bias corpus measurement

Question: Do velocity-biased training corpora show measurably lower IRI scores than velocity-normalised corpora, as predicted by Training Data Velocity Bias?

Method. Construct two corpora from Common Crawl archives (2020-2025): one sampled proportional to social media sharing velocity (velocity-biased), one sampled uniformly or inversely proportional to velocity (velocity-normalised). Using the validated prototype from Experiment 2.3, measure each document's content fidelity against a Tier 2 reference constructed from its own cited or linked primary sources under T1.0, reported as Ia $\pm$ U per Section 41, for a random sample of 1,000 documents from each corpus. Documents lacking traceable primaries are scored on a proxy quantity: the mean mode divergence of each document's canonical rendering against a pooled reference from identifiable primary sources on the same topic, reported as a distribution comparison rather than a certified Ia. Compare distributions.

Predicted outcome. Velocity-biased corpus shows significantly lower mean Ia (predicted difference: 8 to 15 points). Distortion fingerprints show characteristic amplitude reduction on D1 (Provenance) and D4 (Source Plurality) in the velocity-biased sample.

Note. This experiment directly tests VB's central hypothesis using IRI as the measurement instrument. A positive result validates both the VB hypothesis and the IRI instrument simultaneously.

Experiment 3.2: Ω-rate and the VNS governor
Question: Can Ω-rate (the rate of IRI degradation over time) serve as a reliable trigger for VNS-style velocity governance interventions?

Method. Monitor a live information cascade (a viral news event) in real time. At intervals of 1 minute, sample the most widely shared derivative of the original report. Measure IRI scores in near-real-time using the prototype. Compute Ω-rate = d(Ia)/dt. Compare against VNS governor threshold θ. Evaluate whether Ω-rate threshold exceedance predicts subsequent verified misinformation emergence (validated by post-hoc fact-checking within 72 hours).

Predicted outcome. Ω-rate exceeding threshold predicts misinformation emergence with sensitivity $> 70\%$ and specificity $> 60\%$. Mean Ω-rate is significantly more negative for cascades that produce verified misinformation than for cascades that do not.

Experiment 3.3: The κ recovery test
Question: Does substrate decoherence set the grain of the render scale as the channel model predicts, and can the conventionally anchored gauge κ_{conv} be recovered from measured substrate properties by a pre-registered law?

Quantities on the record before comparison. $\kappa_{\mathrm{conv}} = 10/\langle \Delta\Omega \rangle_{\mathrm{A}}$, the convention value from the Ω-90 anchor, with standard uncertainty $\mathrm{u}(\kappa_{\mathrm{conv}})$ propagated from the anchor panel spread, the transducer version, and the Tier 1 reference uncertainty u(R). $\delta\Omega_{\mathrm{min}}$, the measured deviation noise floor. f, the candidate law for κ_{phys}, pre-registered on the thermomagnetic calibration record before κ_{phys} is evaluated.

Method. Stage 1, characterisation. Measure the decoherence rate Γ of the prototype's OAM modes under operating conditions, by fringe-visibility decay as a function of path delay and integration time. Measure the deviation noise floor by repeated null measurements: the same artefact in both arms, deviation distribution recorded across integration times and active mode counts. Fit $\delta\Omega_{\mathrm{min}} = \mathrm{h}(\Gamma\tau_{\mathrm{m}}, \mathrm{N})$.

Stage 2, the grain test. From stage 1, predict the index noise $\mathrm{u}_{\mathrm{inst}}(\mathrm{Ia}) = \kappa_{\mathrm{conv}}\, \delta\Omega_{\mathrm{min}} / \sqrt{\mathrm{N}}$. Compare against the directly observed repeatability of Ia from the Experiment 2.3 repeatability series. Agreement within the stated uncertainty budget confirms that substrate decoherence propagates to the index exactly as the channel model requires. This stage contains no convention content, and it must pass for the instrument model to stand: failure re-enters Phase 2 regardless of what stage 3 would have shown.

Stage 3, the gauge test. Pre-register the candidate law f, motivated by the stage 1 characterisation and stated in dimensionless form, on the calibration record. Compute $\kappa_{\mathrm{phys}} = \mathrm{f}(\Gamma\tau_{\mathrm{m}}, \mathrm{N})$. Compare against κ_{conv}.

Measurements. Γ with uncertainty. The fitted resolution law $\mathrm{h}(\Gamma\tau_{\mathrm{m}}, \mathrm{N})$ with fit quality. Predicted against observed index noise. The ratio $\kappa_{\mathrm{phys}}/\kappa_{\mathrm{conv}}$ with combined uncertainty, reported in all cases.

Decision criteria. Agreement at stage 3 means $|\kappa_{\mathrm{phys}} - \kappa_{\mathrm{conv}}| \leq \mathrm{U_c}$, where $\mathrm{U_c}$ is the combined expanded uncertainty of the pair at coverage factor $\mathrm{k} = 2$. Three verdicts exhaust the outcomes.

DERIVED: stages 2 and 3 both pass. Grain and gauge both trace to the substrate. The native-metrology thesis extends to the scale, and the anchoring convention is retained as the scale's operational definition, now carrying a confirmed physical interpretation.

ANCHORED: stage 2 passes, stage 3 fails. The substrate sets the resolution of the scale; the gauge remains conventional. The IRI holds the epistemic position of Ra and Rf, and holds it in their company. The native-metrology thesis is bounded to the measurand.

INSTRUMENT FAULT: stage 2 fails. Noise propagation contradicts the channel model; the fault reaches deeper than the scale, and the programme returns to Phase 2 before any scale claim is made.

Predicted outcome. Stage 2 is expected to pass; the prediction follows from the same channel mathematics the instrument is built on, and its failure would indict the instrument model itself. Stage 3 is run as a genuine question. The monograph assigns it no advance probability: agreement upgrades the anchoring convention to a checked prediction; disagreement retires the scale extension of the native-metrology thesis and touches nothing else.

Note. This experiment fixes the reach of the native-metrology thesis. Whether light physically realises fidelity loss is settled upstream, in Phase 1. What stage 3 decides is whether the realisation extends from the measurand to the scale: whether the substrate that makes fidelity loss visible also generates the number of index points a deviation costs. A conventional gauge over a physical measurand is the CRI's position, and it has been a respectable metrological position since the CRI's adoption.

Resources. Prototype and apparatus from Phase 2; a calibrated delay line; the Experiment 2.3 repeatability series. Duration: 6 to 8 weeks. Estimated cost: £10,000 to £20,000.

Experiment 3.4: Reference uncertainty coverage

Question: Do the stated Tier 2 uncertainty budgets cover the realised reference error?

The budget of Section 41 makes a claim about itself: that the expanded uncertainty $U = 2u(R_2)$ covers approximately 95% of realised reference errors. The claim is testable, because some events eventually crystallise an authoritative primary record.

Method. Select 30 live events for which an authoritative primary record is expected to publish within 12 months: official inquiry reports, court records, accident investigation reports, post-event official datasets. At event time, construct a Tier 2 reference under the full protocol, state $u(R_2)$ from equation (22), and archive the reference state and its budget on the tape. When the authoritative record publishes, image it as a Tier 1 reference. Measure the archived Tier 2 reference state against the Tier 1 realisation; the measured deviation, in Ω, is the realised reference error for that event. The archived tape makes the trial possible: a reference state recorded at event time remains measurable against a record that publishes months later (Section 14).

Measurements. Realised error per event. Coverage fraction: the proportion of events whose realised error falls within the stated expanded uncertainty. Stratification of coverage by u_{com} class and by n_{eff}.

Predicted outcome. Coverage between 85% and 98% at $k = 2$, with under-coverage concentrated in events flagged at construction time with high u_{com}.

Success criterion. Coverage $\geq$ 85% at $k = 2$. Below this, the stated budgets are falsified as stated. The response: re-evaluate the component magnitudes (u_w and u_{com} are the designed inflation points), recalibrate, and re-test on held-out events. Persistent under-coverage after

one recalibration cycle restricts certified measurement to Tier 1 references and demotes Tier 2 results to indicative status.

Note. This experiment tests the framework's statements about its own uncertainty. A metrology whose stated uncertainties fail coverage carries them as decoration; the coverage trial is the difference.

Resources. Phase 2 prototype. Standing panel over 12 months, run alongside the other Phase 3 experiments. Personnel: 1 research coordinator. Estimated cost: £20,000 to £35,000.

59. Phase 3 summary

Exp.	Tests	Duration	Cost
3.1	Velocity bias corpus measurement	8-12 weeks	£30k-£50k
3.2	Ω-rate and VNS governor	Ongoing (live events)	£20k-£40k
3.3	κ recovery: grain and gauge	6-8 weeks	£10k-£20k
3.4	Reference uncertainty coverage	12 months (standing panel)	£20k-£35k

Phase 3 total: 12 to 20 weeks for Experiments 3.1 to 3.3, with the coverage panel (Experiment 3.4) running 12 months from the availability of the Phase 2 prototype. £80,000 to £145,000.

Phase 3 decision gate: Results determine the strength of the claims that can be made. If 3.1 is positive, VB is validated. If 3.2 is positive, VNS governance is physically grounded. If 3.3 returns DERIVED, the native-metrology thesis extends to the scale; if it returns ANCHORED, the render scale keeps the position every colour fidelity index has held since the CRI's adoption. If 3.4 passes coverage, the uncertainty budgets are validated as stated and Tier 2 certificates carry full standing. Each result is independently valuable.

60. Programme summary

Phase	Tests	Duration	Cost
1: Foundation	Measurand separation, Englert mapping, cascade law, dimensionality	16-24 weeks	£105k-£165k
2: Instrument	OAM encoding, thermomagnetic record, end-to-end prototype	24-32 weeks	£145k-£225k
3: Validation	VB corpus test, VNS governor test, κ recovery, budget coverage	12-20 weeks (+12-month coverage panel)	£80k-£145k

Total programme: 52 to 76 weeks for the core experiments (approximately 12 to 18 months), with all decision gates except 3.4′s coverage verdict closing within this envelope. The coverage panel (Experiment 3.4) requires the Phase 2 prototype and runs 12 months from its availability, extending the full programme to approximately 92 to 108 weeks. £330,000 to £535,000.

This is a realistic budget for a small-team experimental physics programme. For comparison: a single UKRI New Investigator Award provides up to £400,000 over 24 months. The programme is fundable within existing research grant structures.

61. Collaboration

Gizmet Dev Ltd invites experimental collaboration from researchers in quantum optics, photonic computing, information science, and metrology. The theoretical framework, experimental protocols, and all prior papers are published as open access. Correspondence: gizmet@protonmail.com.

Specific capabilities sought: equipped quantum optics laboratory for Phase 1; OAM mode encoding and sorting expertise for Phase 2; access to large-scale web corpus data with velocity metadata for Phase 3.

OPEN QUESTIONS

The framework's claims stop at stated boundaries, and the boundaries are part of the framework. Five non-claims are deliberate. The claim is a shared channel mathematics with a physical realisation; a distorted retelling degrades no physical coherence anywhere, and the framework requires no such event. Experiments 1.2 and 3.3 are assigned to decide whether anything stronger is true. The monograph claims no theorem status for the multiplicative law: the data processing inequality proves monotone loss, the product form is the memoryless model, and Experiment 1.3 can reject the model while the theorem stands. The monograph claims no strict conservation in open interactions: equality in the duality relation is the closed-ledger limit (Section 30). The monograph claims no Landauer cost for observation as such: the k_{BT} ln 2 price attaches to erasure, and its magnitude against the instrument's power budget is conceptual rather than budgetary (Section 33). The monograph claims no derived κ: until Experiment 3.3 reports, the scale rests on the anchoring convention of Part Six.

Three further claims are deliberately withheld, each marking a live research frontier. The first withheld claim is that the optical layer reads meaning. Under T1.0 the physical layer measures divergence between canonical renderings; semantics enters through the declared transducer and through the empirical correlations the experimental programme tests. The second withheld claim is that the magnitude of optical deviation measures epistemic severity. One inverted negation can outweigh a hundred synonym substitutions, and severity is assigned in the dimensional layer. The third withheld claim is that artefact integrity certifies truth. A pristine original of a false document scores perfectly on V_A.

The following questions remain open after the experimental programme. They represent the frontier beyond the current framework.

1. Formal definition of Information Fidelity Space as a metric space. The axioms of Section 31 constrain the index and leave the metric structure undetermined; the geometry may prove non-Euclidean. The axis normalisation retained by convention in Section 16, equation (7), is subsumed by this item.
2. The exact formal relationship between Ia and a mutual-information functional under a declared encoding. Section 32 positions the IRI against Shannon mutual information, quantum fidelity, and Fisher information qualitatively; the quantitative half, whether Ia bounds or is bounded by an information functional under stated channel assumptions, remains open.
3. Empirical uncertainty floors for constructed references. The budget structure is defined (Sections 40 and 41) and the coverage trial is designed (Experiment 3.4). The achievable magnitude of $u(R_2)$ for live events, its variation across domains and media, and the domains in which Tier 2 references remain permanently unfit for purpose are open empirical questions.
4. Cross-cultural validity: does the dimensional structure of information distortion vary across languages, media types, and cultural contexts?
5. PSI full-architecture integration: connecting the IRI measurement capability to the full PSI inference stack, including the critical-path experiments for nonlinear activation and cross-phase modulation.

6. Ledger closure: experimental confirmation that visibility lost by the system is recovered in measured correlations with the path marker and the environment, and that the instrument's erasure costs track the Landauer bound.
7. The gauge derivation. Whether a pre-registered law $\kappa_{\text{phys}} = \text{f}(\Gamma\tau_{\text{m}}, \text{N})$ reproduces the conventional κ_{conv} stays open until Experiment 3.3 stage 3 runs. The monograph claims no outcome. Disagreement would retire the scale extension of the native-metrology thesis and leave the measurand's physical status untouched.
8. The estimator relation. The defining form of the index, equation (6) in fidelity space, and the instrument form, equation (13) in the mode basis, coincide at the anchor by construction. Their relation away from the anchor, including the conditions under which (13) is an unbiased estimator of (6), requires formal treatment.
9. Severity weighting. The physical layer detects and localises divergence; the epistemic weight of a given divergence is assigned in the dimensional layer. Whether any transducer-side severity weighting can satisfy the content-blind requirement is open.
10. Claim-level canonicalisation. T1.0 measures textual fidelity. A T2.x family canonicalising propositions rather than characters would move the measurand towards semantic fidelity at the cost of a heavier and more contestable specification, and until such a transducer exists and passes its own form-invariance control, faithful paraphrase scores as divergence and is declared as such. Whether a claim-level transducer can remain content-blind is the sharpest open problem in the framework.
11. Adversarial reference construction. Fingerprint-based independence analysis has a detection floor: a sufficiently coordinated network of nominally independent sources, paraphrasing a common script, can hold fingerprint correlations below threshold while sharing origin. The detection floor of propagation analysis under adversarial coordination is an open problem, and the honesty of Tier 2 budgets under attack depends on it.
12. IP strategy: IRI as framework, PSI as instrument, each protectable independently.
13. Standardisation pathway: engagement with national measurement institutes (NPL, NIST, BIPM) to evaluate the render (Ω) as a candidate metrological standard.

APPENDIX: DERIVATION

The compounding error across Dionysius, Livy, and Cary; its compression into four-year political cycles; the recognition that information has no metrology; CRI as structural precedent, including its anchoring convention; the Information Rendering Index with two declared measurands and a canonicalisation transducer; unification with PSI; the recognition that light realises information fidelity, and that interferometry has read a fidelity index off a screen since Young; and the uncertainty doctrine that prices every reference.

In retrospect, the same chain connects all three prior papers. Velocity Bias measured the symptom. Vital Network Science proposed the treatment. PSI built the instrument. This monograph identifies the disease.

The chain is the path. The monograph is the destination.

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